



Industrial IIoT Implementation

Bridging Operational Technology with Cloud-Native Edge Computing for Autonomous Manufacturing



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The idea in one page

One secure IIoT + AI platform across utilities, production, furnace/boiler supervision and predictive maintenance.

1 UTILITIES INTELLIGENCE

Electricity, water & gas monitoring
Cost per vehicle / line
Leakage & overconsumption alerts

2 PLC VISIBILITY & CONTROL

Unified PLC signals via edge gateways
Body count, status & downtime
Shift and daily reporting

3 FURNACE & BOILER SUPERVISION

Temperature, pressure & flow watch
Excessive / rapid-rise detection
Alarm + PLC-approved response

4 ML PREDICTIVE MAINTENANCE

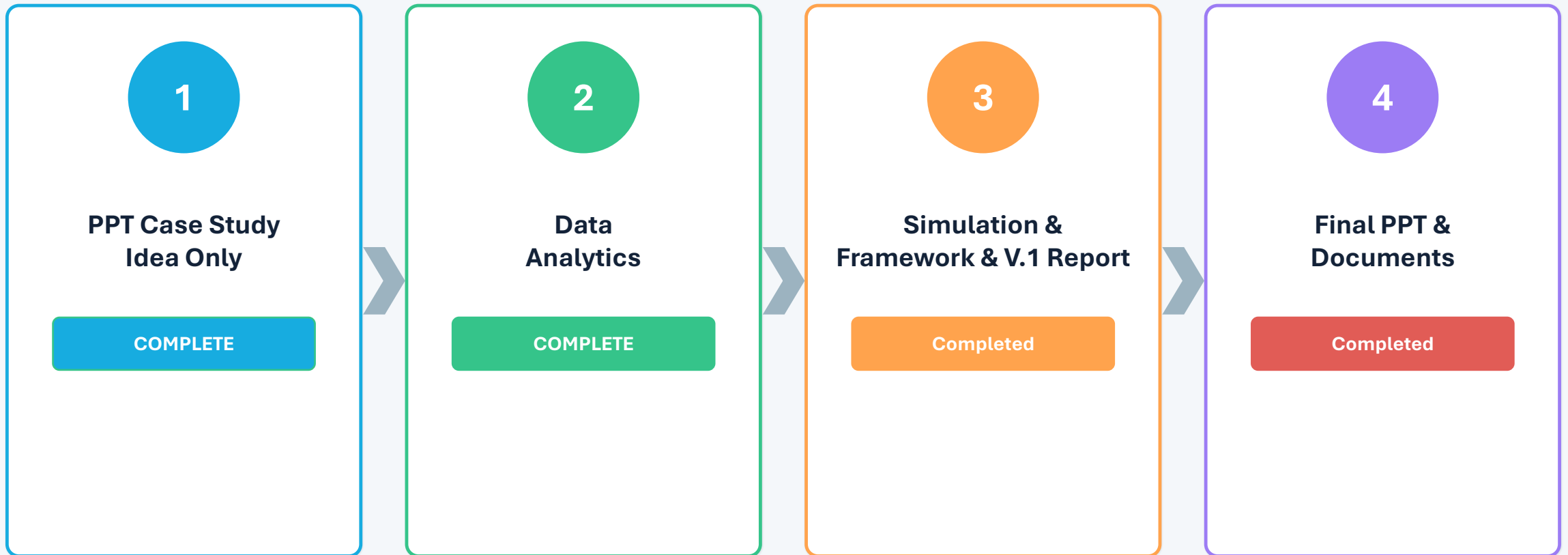
Learns normal machine behavior
Detects anomalies early
Estimates upcoming failure risk

WHY IT MATTERS

Lower utility cost per vehicle produced
Safer, supervised furnace / boiler operation
Less unplanned downtime through early warning
A statistically validated cost model, even where raw sensor data is imperfect

Project roadmap — 4 work tasks

The IIoT case study is delivered through four sequential tasks, each with a clear deliverable and status.



This update covers the status and evidence for each task, followed by the recommended next steps.

PoC 10/10 Activities Complete

Executed by IoT Misr — site execution confirms the case-study concept is technically proven on real plant equipment.

✓ Site survey	✓ AVEVA software install & license activation
✓ Power meter communication test	✓ Flow meter installation & configuration
✓ Gateway installation & configuration	✓ Flow meter → gateway Modbus mapping
✓ Gateway → AVEVA Modbus mapping	✓ Power meter Modbus mapping
✓ Factory PLC communication & IO mapping	✓ Dashboard creation

10/10 Activities completed	3 Data sources integrated	5 Live platform pages
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Field Instrumentation Installed On-Site

Equipment deployed during the PoC, with key parameters sourced directly from the vendor technical datasheets.

WATER



Ultrasonic Flow Meter — FUF-2000B

Transit-time ultrasonic • Accuracy 0.5–1% (Turkak) • DN25–DN6000 • 0–12 m/s • RS485 Modbus

ELECTRICITY



3-Phase Digital Power Meter

Panel-mounted • Per-phase V / I / kWh • CT-based measurement • Modbus to AVEVA

GAS



Vortex Flow Meter — VTRANS Series

Kármán vortex principle • Accuracy 0.5% liquid / 1% gas • 12–32 VDC • 4–20mA+HART / RS485

GATEWAY



Factory PLC & AVEVA HMI

Mitsubishi PLC I/O + AVEVA InTouch • line status, counters, utilities, alarms

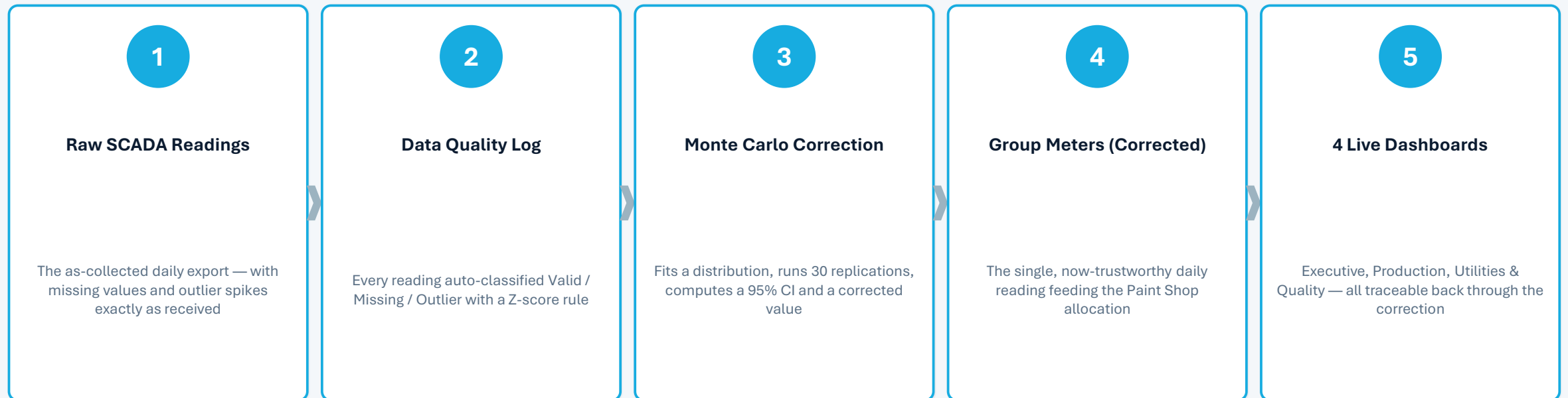
TASK 2 EVIDENCE — THE PAIN POINT

Why This Cost Model Exists

Prima Factory has only ONE meter per utility for the entire manufacturing group — there is no dedicated Paint Shop sub-meter today.

THE PAIN POINT

One electricity meter, one water meter, and one gas meter cover the WHOLE Prima manufacturing group. There is no dedicated Paint Shop sub-meter. On top of that, the raw meter export itself is imperfect — missing readings and outlier spikes occur on several days, so even the one meter we have cannot be trusted at face value.



Until dedicated Paint Shop sub-meters are installed, this workbook is the interim, auditable, statistically-corrected model.

The Raw Data Was Not Clean

A review of the August 2026 SCADA export found real data-quality problems on the single Group meter feed — before any correction.



MISSING READINGS

8 daily readings blank across Electricity / Water / Gas

Cause: sensor / communication dropouts

Example: 05-Aug, 14-Aug, 23-Aug electricity



OUTLIER SPIKES

8 daily readings deviating $>2.5\sigma$ from normal

Cause: comms glitches / signal noise

Example: 08-Aug electricity read 2,561 kWh vs. ~13,300 kWh expected



WHY IT MATTERS

16 of 83 readings (19%) needed correction

Left uncorrected, monthly totals would be biased

A defensible fix needs a documented statistical method, not manual edits

Reviewed with a Data Quality Log using an automatic Z-score classification rule, per every reading, per utility.

Monte Carlo Simulation — How the Data Was Corrected

The correction technique follows the course's Input Data Modeling & Output Analysis pipeline: Collect → Clean → Explore → Fit → Test → Apply.

1**FIT**

For each flagged (missing / outlier) reading, fit a Normal distribution from the valid readings of the same utility and same day-type (weekday / weekend).

2**REPLICATE**

Draw $N = 30$ independent replications from the fitted distribution — the course's recommended minimum for a reliable sample (10–30 replications).

3**QUANTIFY**

Compute the sample mean, sample standard deviation, standard error ($\sigma/\sqrt{N}$), and a 95% confidence interval ($\text{mean} \pm 1.96 \times \text{SE}$) for every corrected value.

4**APPLY**

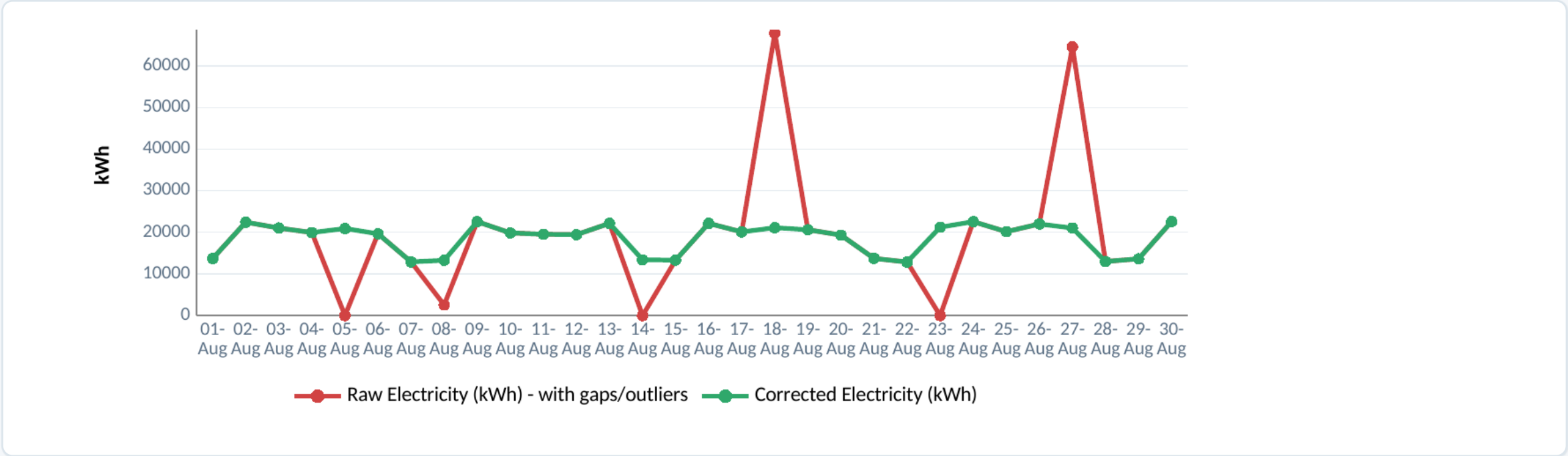
Replace the flagged reading with the replication sample mean; valid readings pass through unchanged into the corrected Group Meter series.

Reference: Simulation Modeling course — Week 6 Input Data Modeling (Collect→Clean→Explore→Fit→Test→Apply) and Output Analysis (Replication, Standard Error, 95% Confidence Interval).

TASK 3 EVIDENCE — THE CORRECTION IN ACTION

Raw (Noisy) vs Monte-Carlo-Corrected Electricity

August 2026 daily Group meter electricity reading, before and after correction. Red drops to zero = missing; red spikes = outliers; green = corrected.



6

ELECTRICITY READINGS CORRECTED

3.6%

AVG. CORRECTION ERROR VS. EXPECTED PATTERN

309,115

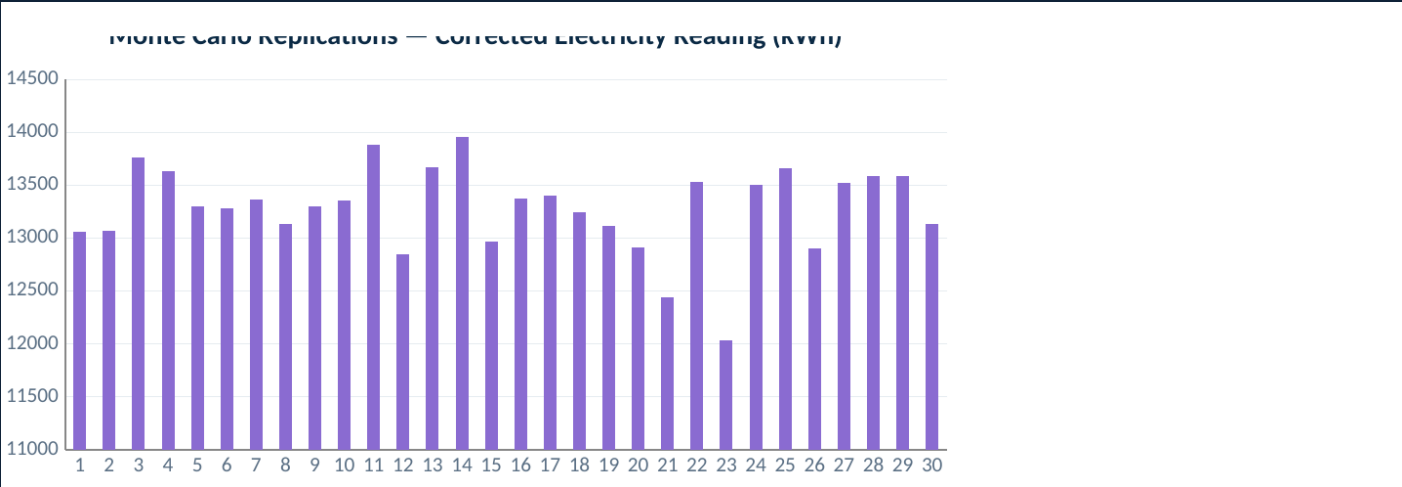
CORRECTED MONTHLY COST TOTAL (EGP)

0.04%

SHIFT VS. UNCORRECTED BASELINE TOTAL

Inside One Correction — 30 Monte Carlo Replications

Worked example: the 08-Aug-2026 electricity reading was flagged as an outlier (2,561 kWh vs. ~13,300 kWh expected). Here is how it was corrected.



REPLICATION STATISTICS

Fitted μ (from valid days)	13,300.3 kWh
Fitted σ	389.2 kWh
Sample Mean (N=30)	13,280.18 kWh
Sample Std Dev	408.41 kWh
Standard Error	74.57 kWh
95% CI	[13,134.03 , 13,426.33]
Corrected Value	13,280.18 kWh

Verification & Validation of the Correction

"Did we build the model right?" (Verification) and "Did we build the right model?" (Validation) — per the course V&V framework.

VERIFICATION

Built the model right?

Every flagged cell traced to a Monte Carlo formula, not a manual edit
Corrected value = AVERAGE of 30 live replications (auditable in-sheet)
Valid readings pass through unchanged — spot-checked against raw export

VALIDATION

Built the right model?

Corrected series compared against the expected same-day-type pattern
Avg. correction error: Electricity 3.6% · Water 6.4% · Gas 2.5%
Monthly total shifts only 0.04% vs. an uncorrected reference baseline

Conclusion: the correction is both verified (correctly implemented) and validated (results close to the expected pattern) — defensible for management reporting.

Cost to Paint One Vehicle — Post-Correction

Formula-linked totals for all 2,200 vehicles produced in August 2026, computed from the Monte-Carlo-corrected Group meter.

2,200

VEHICLES PAINTED

309,115

TOTAL PAINTING COST (EGP)

140.51

AVG COST / VEHICLE (EGP)

16 of 83

READINGS CORRECTED VIA MONTE CARLO

19.3%

% REQUIRING CORRECTION

GB_Auto_Prima_Factory_Dataset.xlsx — Home

GB AUTO — PRIMA FACTORY PAINT SHOP UTILITY COST MODEL

Cost of painting one vehicle — electricity, water and gas (Monte-Carlo data-corrected) | Data as of

EXECUTIVE DASHBOARD →

PRODUCTION DASHBOARD →

UTILITIES DASHBOARD →

QUALITY DASHBOARD →

MONTE CARLO CORRECTION →

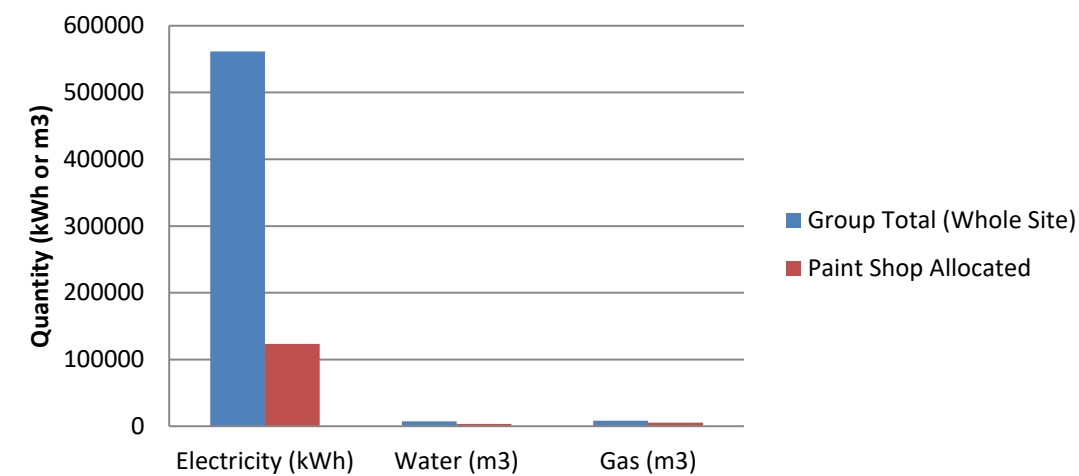
DATA QUALITY LOG →

HEADLINE KPIS

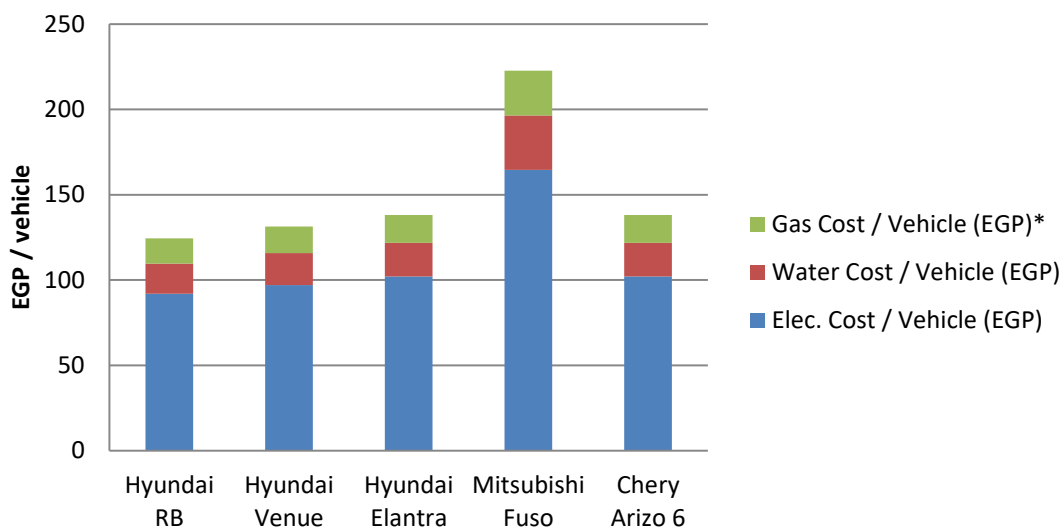
VEHICLES PAINTED	TOTAL PAINTING COST (EGP)	AVG COST / VEHICLE (EGP)	ELEC. COST / VEHICLE (EGP)	EGS CORRECTED VIA MONTE
2,200	309,115	140.51	103.60	16

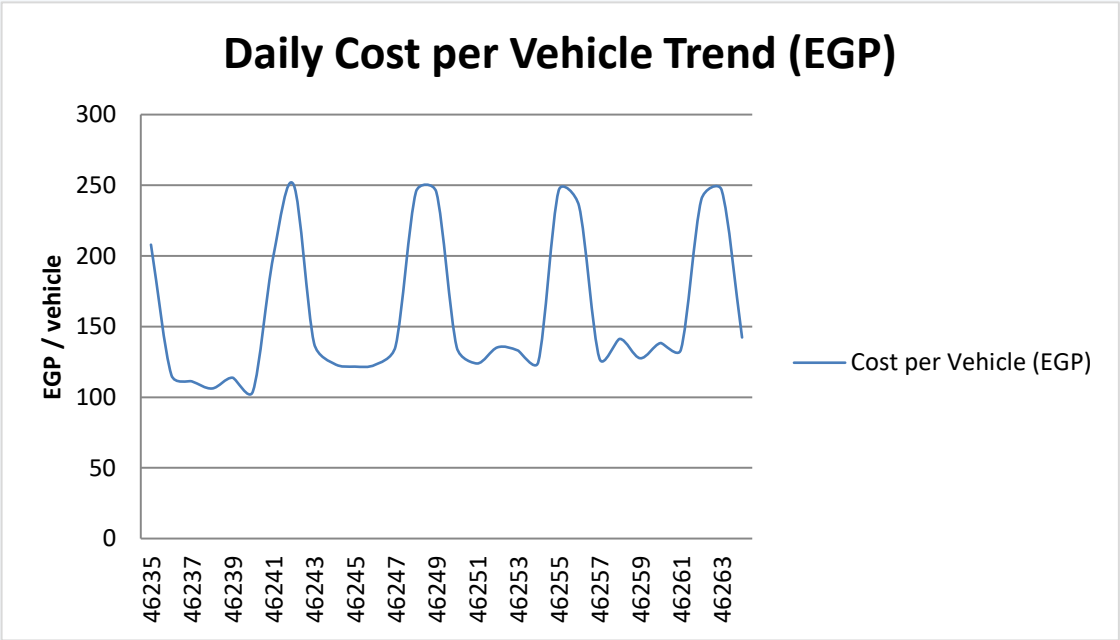
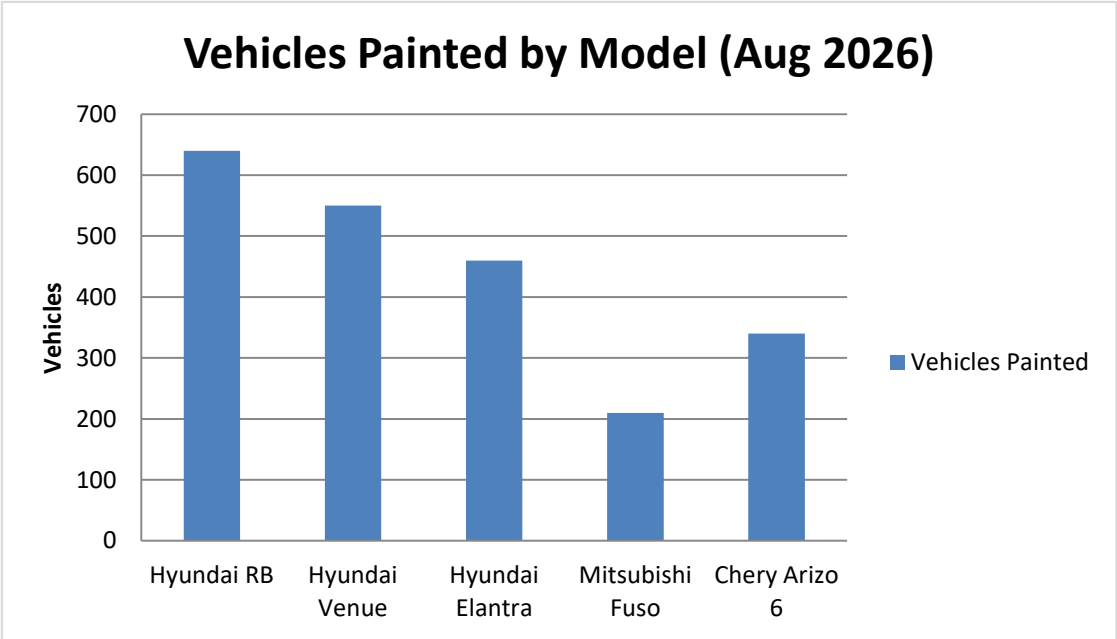
Paint shop model coverage: Hyundai RB · Hyundai Venue · Hyundai Elantra · Mitsubishi Fuso · Chery Arizo 6

Group Meter Total vs Paint Shop Allocated
(August 2026)



Cost to Paint One Vehicle, by Model (EGP)





TASK 3 EVIDENCE

Framework and Projected ROI

The end-to-end framework is defined; IoT Misr's PoC-based simulation quantifies the expected operational and financial return.

PRODUCTION KPI IMPACT

25–35%

Projected impact on production KPIs

kWh / vehicle benchmarking
Line-to-line efficiency comparison

OPERATOR & ENGINEERING EFFICIENCY

25–35%

Projected reduction in operational effort

Faster anomaly detection
30–40% less engineering effort
Automatic reporting

ENERGY CONSUMPTION REDUCTION

~12%

Weighted average total reduction

Electricity 8–12%
Natural gas 10–15%
Water 5–10%

Basis: conservative, achievable ranges from the PoC design, as reported by IoT Misr — verified verbatim against the "Operation & Production ROI" section of the vendor PoC deck.

Thank You